

The HBC Single-cell Course, in BioLang

The Harvard Chan Bioinformatics Core teaches a fourteen-lesson introduction to single-cell RNA-seq. It is one of the better free curricula in the field, it is taught in R with Seurat, and it is released under CC BY 4.0.

This book walks the same road in BioLang, on the same data.

Why follow someone else's syllabus

Because the ordering is the hard part, and theirs is good.

A tool's documentation organises itself around the tool: here are the functions, here is what each takes. A course organises itself around the learner: here is what you cannot understand until you understand this other thing first. HBC puts the theory of PCA *before* normalization, which looks backwards until you notice you cannot judge whether a normalization worked without a way to look at the result. They give cluster quality control an entire lesson, long after clustering, because "are these clusters real?" only becomes answerable once you have some.

Why on their data

An earlier version of this material ran on a small synthetic fixture. It was fast, it needed no download, and it was wrong in a way that mattered: the genes were named `MARK0_000`, `MARK1_000`, and so on. Every mechanical step worked, and the one thing the annotation lesson exists to teach — looking at `LYZ`, `CD14` and `S100A8` together and concluding *monocytes* — could not be demonstrated at all.

So this book uses the course's own PBMC dataset. It costs a download and some runtime. In exchange, the figures show recognisable immune biology, and every number can be compared against what the course reports.

It also surfaced things a tidy fixture never would. Real clusters range from 3,128 cells to 32, and that unevenness alone found a latent crash in the violin plots. Real raw matrices have 737,280 droplets, which found a plotting path that never terminated. Synthetic data tests the code you wrote; real data tests the assumptions you did not know you had made.

What you need

BioLang, the `singlecell` package, and about 90 MB of extracted matrices from a 3.2 GB archive. All three steps are in Getting the Data.

What this is not

It is not a replacement for the course. Read the HBC lessons for the biology and the reasoning; read these chapters when you want to run the same steps here. It is also not a claim that BioLang reproduces Seurat exactly — it does not, and What Differs from the Course is specific about where and why.

Credit, licence, and the full list of changes are on Attribution and Licence. Read that first if you plan to reuse any of this.

A note on the code

None of it runs in the browser. Every example imports a package and most read or write files, and package imports and file I/O are CLI-only — so these pages show no Run button. Copy the code, or take the scripts and notebook from Downloads.

Attribution and licence

The source

This book follows the curriculum of:

Introduction to Single-cell RNA-seq Harvard Chan Bioinformatics Core (HBC) Mary Piper, Meeta Mistry, Radhika Khetani, Lorena Pantano, Jihe Liu, Will Gammerdinger, and Noor Sohail <https://hbctraining.github.io/Intro-to-scRNAseq/>

The teaching sequence is theirs, and it is the most valuable thing here: the decision to explain PCA before normalization, to separate the theory of integration from the code that runs it, to give cluster quality control a lesson of its own rather than a paragraph, and to close with the whole workflow written out end to end.

It runs on **their dataset** as well — the PBMC control/interferon-stimulated experiment their course uses — so the numbers on these pages can be compared against theirs directly. That is the point of this book: it is a translation, not a re-illustration.

If this companion is useful, the course it follows is more so. It is free to read, and it is taught in R with Seurat, which remains the reference implementation that any other tool is measured against.

Licence of the source

The HBC lessons are released under **Creative Commons Attribution 4.0 International (CC BY 4.0)**, declared in the front matter of each lesson:

```
license: "CC-BY-4.0"
```

The licence text is at <https://creativecommons.org/licenses/by/4.0/>.

CC BY 4.0 permits adaptation, including commercially, provided credit is given, the licence is linked, and **changes are indicated**. This page is that indication.

What was changed

- **The language.** The course teaches R with Seurat; this book teaches BioLang. No code here is a line-by-line translation of theirs.
- **All prose is written from scratch.** No text is copied. Where a passage explains the same idea, it explains it in different words.
- **No figures are reproduced.** Several of the course's are credited to third parties (Trapnell 2015, Wagner 2016, Hicks 2015) whose licences would each need tracing. Every figure here is generated by the code on the page that shows it.
- **No data is redistributed.** Getting the Data downloads it from the authors' own host, which is what their `download.sh` does.
- **Fourteen lessons are regrouped into seven chapters.** Some course lessons are R environment setup that BioLang does not need. The mapping, and the places where BioLang cannot follow, are in What Differs from the Course.
- **The numbers will not match exactly.** Same data and same thresholds, but different implementations of HVG selection, PCA and community detection give different cluster counts and boundaries. Where a result should agree, this book says so and shows it; where it will not, it says that too.

Licence of this book

These chapters are part of the BioLang repository and carry its MIT licence, covering the prose and BioLang code written here.

Because this adapts a CC BY 4.0 work, the attribution requirement travels with it: if you reuse these chapters, keep the credit to the HBC training team and the link to their course.

Citing

Cite the course this follows:

```
@misc{piper2022scrnaseq,
  title = {hbctraining/scRNA-seq_online: scRNA-seq Lessons from HCBC},
  author = {Piper, Mary and Mistry, Meeta and Liu, Jihe and
    Gammerdinger, William and Khetani, Radhika},
  year = {2022},
  month = {1},
  publisher = {Zenodo},
  doi = {10.5281/zenodo.5826256},
  url = {https://doi.org/10.5281/zenodo.5826256}
}
```

The HBC team asks for citations directly, and gives the reason: they show the community's needs, earn recognition for the work, and support further funding for their teaching. That is a good reason, and it costs you one line.

The data

The PBMC experiment is from Kang et al., *Nature Biotechnology* 2018 — peripheral blood mononuclear cells from lupus patients, split into a control sample and one stimulated with interferon-beta. The HBC course hosts a packaged copy; this book downloads that copy and does not redistribute it.

Getting the data

Everything in this book runs on the HBC course's own dataset. Three steps, once.

1. Install BioLang

Linux / macOS

```
curl -fsSL https://lang.bio/install.sh | sh
```

Windows (PowerShell)

```
iwr -useb https://lang.bio/install.ps1 | iex
```

Check it:

```
bl --version
```

2. Install the singlecell package

Download **singlecell-starter.zip**, unzip it, and from inside:

```
bl install ./singlecell
```

There is no package registry yet, and `bl install --git <url>` clones a whole repository into the package slot — which cannot work for a package living in a subdirectory of one. Installing from a local path is the supported route, so the kit ships the package rather than pointing at it.

3. Download the matrices

```
python get-data.py
```

Be warned about the size. The archive is **3.2 GB**; the two directories this book needs total about **90 MB**. Zip keeps its index at the end of the file, so there is no way to pull two members out of a remote archive without fetching the whole thing — HBC's own `download.sh` has the same problem. The script downloads once, extracts what is used, and deletes the archive, so the 3.2 GB is transient rather than resident. Pass `--keep` to hold on to it.

The rest of the archive is R objects — a 2.2 GB `seurat_integrated.RData.bz2` among them — that BioLang cannot read and this book does not need.

When it finishes you have:

```
ctrl_raw/      barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
stim_raw/      barcodes.tsv.gz  features.tsv.gz  matrix.mtx.gz
```

Run everything from the directory containing those.

Check it worked

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let ctrl = sc.load("ctrl_raw")
println("droplets: " + str(ctrl.n_cells))
println("genes:    " + str(ctrl.n_genes))
```

which prints:

```
droplets: 737280
genes:    33538
```

737,280 is every barcode the chemistry can produce, not every cell — the vast majority are empty droplets. Sorting that out is Quality Control.

What the experiment is

Peripheral blood mononuclear cells from lupus patients (Kang et al., *Nature Biotechnology* 2018), in two samples:

Sample	
ctrl_raw	untreated control
stim_raw	stimulated with interferon-beta

Two samples of the same cells under different conditions is what makes this dataset worth teaching on: it has real biological differences *and* a real batch structure, so Integration has something honest to work against rather than a planted offset.

A note on time

This is a real dataset and BioLang runs it single-threaded. On a laptop:

Step	Roughly
Load one raw matrix	2 s
Filter to cells	5 s
Normalize, HVG, PCA, neighbours, cluster	15 s
The same on both samples merged	40 s
Harmony on 30,000 cells	minutes

Nothing here needs a cluster, but the two-sample chapters are not instant.

The Biology and the Matrix

Follows HBC lessons 01 (Intro to scRNA-seq) and 02 (Generation of the count matrix).

What the experiment buys you

Bulk RNA-seq measures an average. Grind up a tissue, sequence the RNA, and you learn what the average cell was transcribing. The average is a real number, and it is often the wrong one.

The classic failure: suppose genes A and B are positively correlated *within* every cell type, but the cell types sit at different overall levels. Average across the mixture and the correlation can invert. You will confidently report that A suppresses B when in every actual cell it does the opposite. This is Simpson's paradox with a pipette, and it is not an edge case — it is the default hazard whenever a tissue contains more than one kind of cell, which is every tissue.

Single-cell RNA-seq measures cells separately, so the grouping the average destroyed is still there to recover. That buys you which cell types are present and in what proportion, rare populations a bulk average dilutes away, how expression shifts along a differentiation trajectory, and — the one that changes conclusions most often — differential expression **within a cell type** between conditions. “Gene X is up in disease” is a different claim from “gene X is up in the monocytes of treated patients”, and only the second tells you where to look next.

That last distinction is exactly what this dataset is built for: the same patients' cells, control and interferon-stimulated.

What it costs you

The data is large. This one is 33,538 genes by 737,280 droplets per sample. Stored densely that is 24 billion entries; stored sparsely it is about 10 million.

`sc.load` returns a sparse matrix for that reason.

Sequencing per cell is shallow. Droplet methods detect perhaps 10–50% of the transcriptome in a given cell, so a zero is ambiguous: the gene was off, or it was on and you missed it. You cannot tell which from the number.

This ambiguity is the single most important fact about the data type. The vocabulary is worth being careful about: scRNA-seq data is often called *zero-inflated*, meaning more zeros than a count model predicts. Recent analyses argue it mostly is not — the zeros are about what sequencing depth would predict, and a plain negative binomial handles them. They are real dropouts, not evidence of a second process. This matters because “zero-inflated” invites you to add machinery for it, and the machinery can do harm.

Biological variation you did not ask about. Transcription is bursty. Cells cycle. Cells respond to neighbours. Identity is sometimes a gradient rather than a category. All real biology, none of it the biology you are studying, all of it in the same matrix.

Technical variation. Capture efficiency differs per cell, amplification is not uniform, libraries differ in quality, and batches differ from each other — sometimes more than the conditions do. That is why integration gets two lessons.

Where the matrix comes from

BioLang starts one step downstream of the sequencer. Worth knowing what that step did.

In a droplet experiment each droplet ideally captures one cell and one bead. The bead carries millions of oligos sharing a **cell barcode**, so everything sequenced from that droplet is stamped with the same identifier. Each oligo also carries a **unique molecular identifier (UMI)**, a short random sequence differing between oligos on the same bead.

The UMI is the clever part. PCR amplifies some molecules more than others, so read counts are a distorted measure of how much RNA was there. But two reads sharing a UMI *and* a cell barcode *and* a gene came from one original molecule. Collapse them, count distinct UMIs, and you have counted molecules instead of reads. Amplification bias largely disappears.

Cell Ranger does the demultiplexing, barcode correction, alignment and UMI collapsing, and writes three files: `barcodes.tsv` (the columns), `features.tsv` (the rows) and `matrix.mtx` (the non-zero counts).

BioLang does not run Cell Ranger, and neither does Seurat. That work is upstream of both. What BioLang does is read its output:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let ctrl = sc.load("ctrl_raw")

println("droplets: " + str(ctrl.n_cells))
println("genes:      " + str(ctrl.n_genes))
println("first genes: " + str(sc.get_genes(ctrl) |> take(5)))
```

```
droplets: 737280
genes:      33538
first genes: [MIR1302-2HG, FAM138A, OR4F5, AL627309.1, AL627309.3]
```

Two things to notice. The gene names are real symbols, which is what makes annotation possible later. And 737,280 is every barcode the chemistry can produce — the overwhelming majority are empty droplets that caught only ambient RNA. The matrix as loaded is mostly nothing.

The object is a plain BioLang record — `matrix`, `genes`, `barcodes`, `n_cells`, `n_genes`. Later steps add fields rather than replacing it, so `keys(obj)` always tells you what has happened.

What a count actually means

One entry is: *the number of distinct mRNA molecules from gene G that were captured, reverse transcribed, amplified, sequenced, and successfully assigned to barcode C.*

Every verb can fail, and each fails at a rate that varies between cells. Holding onto that is what stops you over-reading a small number later. A count of zero is weak evidence of absence. A count of one is weak evidence of anything.

Next

Separating the cells from the empty droplets is Quality Control.

Quality Control

Follows HBC lessons 03 (QC setup), 04 (Cell Ranger QC), and 05 (Quality control).

The thing being filtered is not a cell

The matrix has one column per barcode, and the course is careful to say *barcode* rather than *cell*, because 737,280 of them are not cells:

- **Empty droplets** that caught only ambient RNA — transcripts from cells that lysed during dissociation. Few genes, low counts, and the vast majority of the matrix.
- **Dying cells**, which leak cytoplasmic RNA while mitochondria stay intact longer, so the mitochondrial fraction climbs as the cytoplasm drains.
- **Doublets** — two cells in one droplet, appearing as one barcode with roughly twice the content and an impossible hybrid identity.

Look before you cut

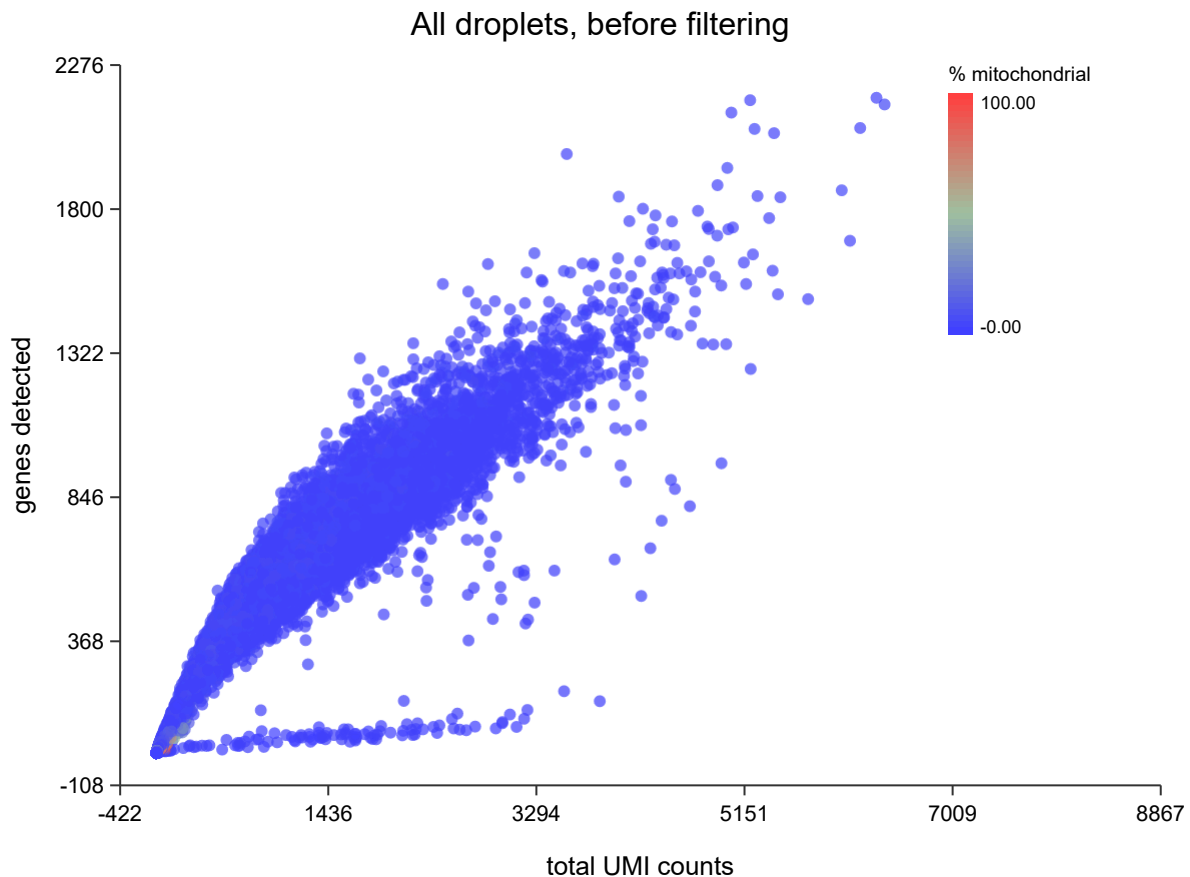
`sc.qc` computes per-cell and per-gene metrics and attaches them without removing anything.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("ctrl_raw")
let scored = sc.qc(raw)

write_text("qc-raw.svg",
  sc.plot_qc_scatter(scored, "All droplets, before filtering"))
```



That is every barcode in the run. Two features carry the whole decision.

The **dense wedge running up and to the right** is the real cells: as a droplet captures more UMIs it detects more distinct genes, and the two rise together.

The **flat spur along the bottom** is the problem. Those droplets accumulate UMIs without accumulating genes — the signature of low-complexity content: ambient RNA, dying cells, or a population dominated by a handful of transcripts. Neither histogram shows this on its own. A cell with few genes *and* few UMIs is an empty droplet; few genes but many UMIs is a different animal entirely, and only the joint view distinguishes them.

The enormous blob at the origin is the empty droplets, and it is most of the 737,280.

The three metrics

Metric	Low means	High means
UMIs per cell	empty droplet, or a small cell	doublet, or a large cell
Genes per cell	empty droplet, low complexity	doublet

Metric	Low means	High means
% mitochondrial	usually fine	dying or stressed cell

Note the second column. **Every one of these has a benign explanation.** A plasma cell genuinely has fewer distinct genes because it is devoting itself to antibody transcripts. Cardiomyocytes are genuinely full of mitochondria. A threshold right for PBMCs will delete a real population in heart tissue. This is why the course insists on plotting distributions rather than applying remembered numbers.

Applying the cuts

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("ctrl_raw")
let clean = raw
  |> sc.filter_genes(3)
  |> sc.filter_cells(250, 100000, 20.0)

println("before: " + str(raw.n_cells) + " droplets, " + str(raw.n_genes) + "
genes")
println("after: " + str(clean.n_cells) + " cells, " + str(clean.n_genes) + "
genes")
```

```
before: 737280 droplets, 33538 genes
after: 15049 cells, 15576 genes
```

737,280 barcodes down to 15,049 cells, and 33,538 genes to 15,576.

That is close to the course's result but **not the same filter**, and the difference is worth being exact about. Theirs has four criteria:

```
nUMI >= 500 & nGene >= 250 & log10GenesPerUMI > 0.80 & mitoRatio < 0.20
```

`sc.filter_cells` expresses two of them — a gene floor and a mitochondrial cap. It has no UMI floor and no complexity term. Applying all four by hand on this sample gives **14,847 cells** against this chapter's **15,049**: a difference of 202 cells, or 1.3%.

The 202 are mostly low-complexity droplets — high UMI counts spread over few genes, the flat spur along the bottom of the plot above. The complexity term `log10GenesPerUMI > 0.80` is what removes them, and it is the criterion worth adding by hand if you are reproducing the course exactly:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("ctrl_raw")
let m = cell_qc(raw.matrix, raw.genes)
let genes = col(m, "n_genes")
let umis = col(m, "total_counts")
let mito = col(m, "pct_mito")

let keep = range(0, raw.n_cells) |> filter(|i| {
  umis[i] >= 500.0 and genes[i] >= 250.0 and mito[i] < 20.0 and
  (log10(genes[i]) / log10(umis[i])) > 0.80
})
println("cells: " + str(len(keep)))
```

```
cells: 14847
```

The chapters that follow use the two-criterion form, because it is the one the package exposes and the 1.3% does not change any conclusion here. If you need the course's numbers exactly, use the block above.

Two filters, in this order and for a reason. `filter_genes(3)` drops genes seen in fewer than three cells — a gene detected once cannot support a statistical claim, and carrying eighteen thousand such genes costs memory and inflates the multiple-testing burden later. `filter_cells(min_genes, max_genes, max_pct_mito)` drops barcodes: the lower bound removes empty droplets, the upper catches doublets, the mitochondrial cap removes dying cells.

The thresholds here are the course's: at least 250 genes, mitochondrial fraction under 20%. The upper gene bound is deliberately loose because the lower bound and the mitochondrial cap are doing the work on this dataset.

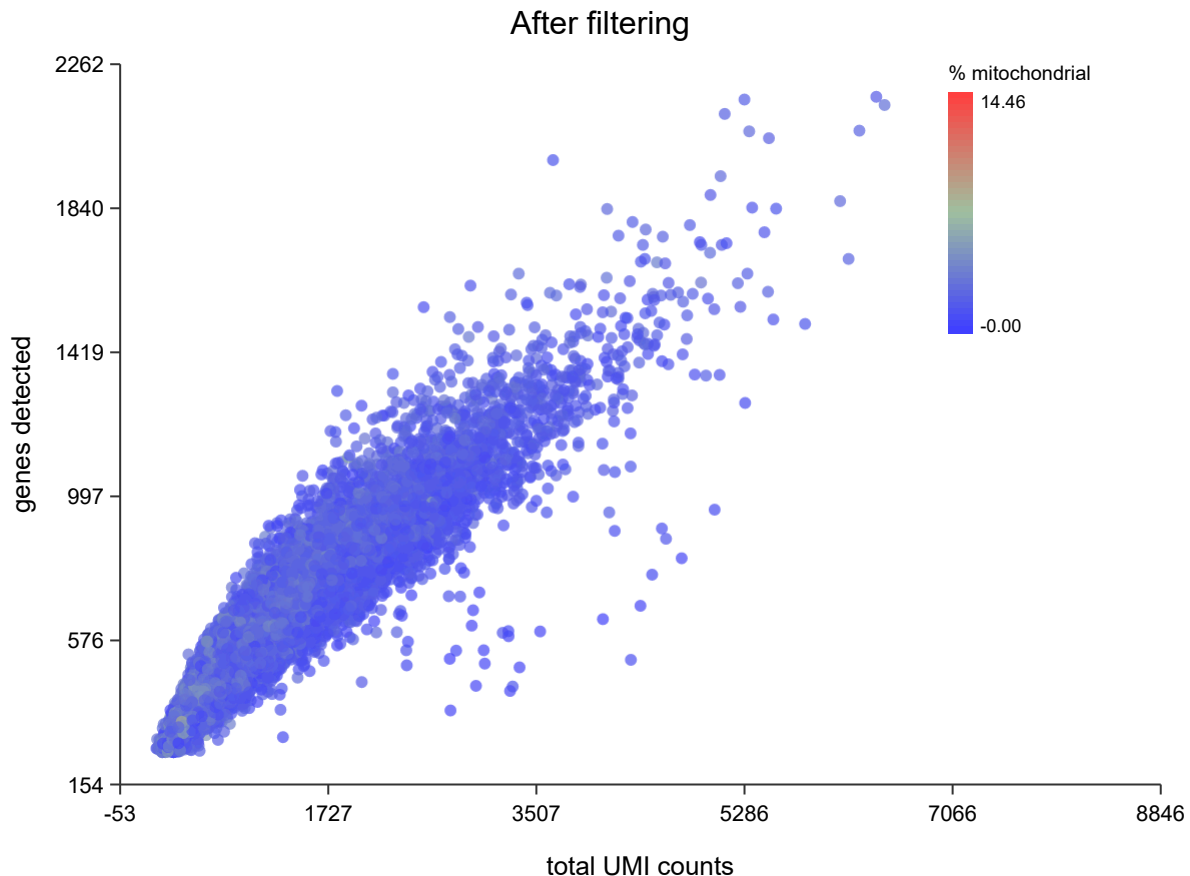
What survived

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let clean = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)

write_text("qc-clean.svg", sc.plot_qc_scatter(sc.qc(clean), "After
filtering"))
```



The spur along the bottom is gone and the blob at the origin with it. What remains is the diagonal band, which is what a clean sample looks like.

The mitochondrial cap needs gene symbols

`max_pct_mito` finds genes whose symbol starts with `MT-`. If your features file carries Ensembl IDs (`ENSG00000198804`) rather than symbols, nothing matches, the mitochondrial fraction is zero for every cell, and the filter silently does nothing.

It will not warn you. This dataset's `features.tsv` has three columns — Ensembl ID, symbol, feature type — and `sc.load` takes the symbol, so the cap works here. Check that your gene names look like names.

What Cell Ranger knew that you no longer do

The course spends a lesson on Cell Ranger's `web_summary.html`. **BioLang cannot read that file**, and most of what it reports can be recomputed from the matrix anyway — cell counts, median genes per cell, UMI distributions.

Two things cannot. **Sequencing saturation** (how much another lane would buy) and **fraction of reads mapped to the transcriptome** are properties of the reads, and the reads are gone by the time you have a matrix. If those matter — and for judging whether an experiment was deep enough, they do — read them in Cell Ranger's viewer before moving on.

Judge the filter by what survives

The only real test of a threshold is whether the biology is still there afterwards. Filter, cluster, and check you still have the populations you expected; if a cell type vanished between raw and filtered, the threshold found it, not the noise.

That check needs clusters, which is Normalization and PCA and then Clustering. QC is not finished until you have been round that loop once.

Normalization and PCA

Follows HBC lessons 06 (Theory of PCA) and 07 (SCTransform normalization).

Why the course teaches PCA first

This looks wrong the first time. Normalization comes before PCA in the pipeline, so why teach PCA first?

Because you cannot judge a normalization without a way to look at its result. “Did this normalization work?” means “are cells now grouped by biology rather than by sequencing depth?”, and answering it requires a projection. Teaching the tool before the step that needs it is the right way round.

PCA in one paragraph

You have a cell described by 15,576 gene measurements — a point in 15,576 dimensions. Most of those dimensions carry nothing: genes nobody expresses, genes everybody expresses equally, genes that are noise. Many move together, because genes work in programs.

PCA finds new axes as weighted combinations of genes, ordered so the first captures the most variance, the second the most of what remains. The first ten or twenty usually carry the structure; the rest carry noise.

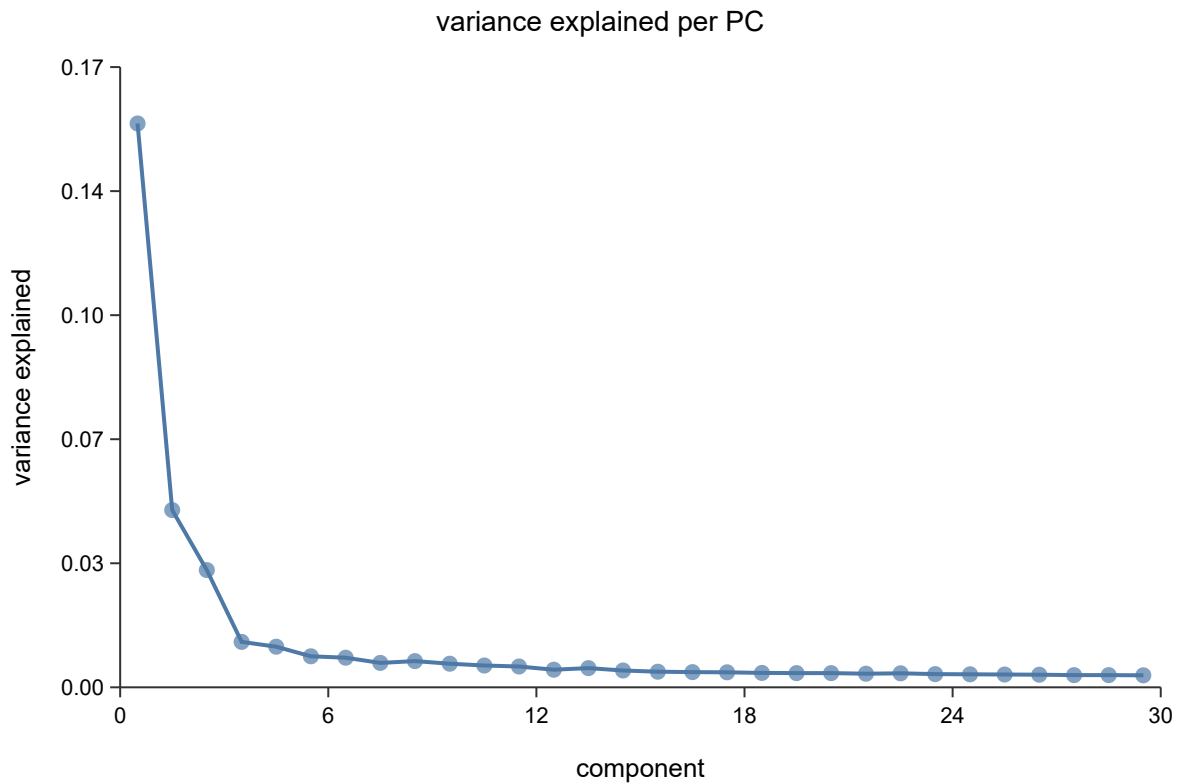
The intuition to keep: **each PC is a gene program**, and a cell’s coordinate on it is how strongly that cell runs the program. PC1 is often “which broad lineage is this”, and later PCs get more specific.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize()
|> sc.variable_genes(2000)
|> sc.run_pca(30)

write_text("elbow.svg", sc.plot_elbow(obj, 30))
```

Read where it flattens. Components before the bend carry structure; after it, noise. The bend is rarely sharp, and that is fine — taking a few too many PCs costs much less than cutting into real signal, so err high. The course settles on a similar range for this data.

Why raw counts cannot go into PCA

Two problems, and they compound.

Depth. One cell yields 20,000 UMIs, its neighbour 5,000. Every gene in the first looks four times higher, and PC1 becomes “how deeply was this cell sequenced” — a technical axis dominating the plot.

Mean-variance coupling. Count variance grows with the mean, so highly expressed genes vary more in absolute terms simply because they are high. PCA maximises variance, so without correction it selects for expression level rather than information.

The standard fix handles both — scale each cell to a common total, then `log1p`:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize(10000.0)

println("normalised layer present: " + str(contains(keys(obj),
"norm_matrix")))
```

`normalize(10000.0)` is counts-per-ten-thousand followed by `log1p`. The 10,000 is arbitrary and does not matter; the log does the work, compressing the high end so a handful of loud genes stop dominating.

Note it **adds** `norm_matrix` rather than overwriting `matrix`. Raw counts stay available, which matters because differential expression should use counts, not logs.

Variable genes

Most genes are uninformative for distinguishing cell types — off everywhere or on everywhere. Selecting genes that vary more than expected *for their expression level* focuses PCA on structure.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize()
|> sc.variable_genes(2000)

println("HVGs: " + str(len(sc.get_hvg_genes(obj))))
```

“More than expected for their expression level” is the important clause. A raw variance ranking returns the highest-expressed genes, which you already knew about. The selection bins genes by mean expression and ranks dispersion *within* each bin, so a moderately expressed gene that switches cleanly between cell types can outrank a loud constitutive one.

2,000 is the conventional number and what the course uses.

SCTransform

Log-normalization has a known weakness: it does not fully remove the depth-expression relationship, and residual depth structure leaks into the PCs. SCTransform fits a regularized negative binomial per gene with sequencing depth as a covariate and uses the Pearson residuals as corrected values.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.sctransform()

println("sctransform applied to " + str(obj.n_cells) + " cells")
```

Use it when depth varies a lot across cells, or when integrating samples sequenced to different depths. Use plain `normalize` when you want something simple, fast and easy to explain.

It costs memory that log-normalization does not

A Pearson residual is $(\text{count} - \mu) / \sqrt{\text{var}}$, and when the count is zero that is $-\mu/\sqrt{\text{var}}$, which is not zero. **So the output is dense no matter how sparse the input was.** A $15,000 \times 15,576$ matrix that occupies a few hundred megabytes as counts becomes 1.9 GB as residuals; the integrated object in Integration becomes 4 GB. This is the step that runs people out of memory, and it is not a bug — it is what the method returns.

The next step throws most of it away. Pass a feature count and only the genes that survive selection are computed:

#11 [▶ Run](#) [▶▶ Run All Above + This](#)

```
let obj = merged |> sc.sctransform(3000)
```

Genes are ranked by the variance of their residuals and the top 3,000 kept, which is `SCTransform(variable.features.n = 3000)` in Seurat and drops that 4 GB to 711 MB. The residual values are identical either way — the argument changes which columns you get back, not what is in them.

A capped call also fills `obj.hvg`, because ranking by residual variance *is* variable-gene selection under this normalization. Do not follow it with `sc.variable_genes`.

The two produce different downstream clusters. Pick one and keep it for the whole analysis rather than switching partway — and say which you used, because a reader cannot tell from the figures.

Next

You now have cells positioned in a space where distance means something. This dataset has two samples, and that space is probably still organised partly by sample rather than by biology — which is Integration.

Integration

Follows HBC lessons 08 (CCA theory) and 09 (Harmony).

The course splits integration into two lessons — one for the theory, one for the code — and the split is worth keeping, because this is the step where a plausible-looking result is most likely to be wrong.

The problem, on this data

Merge the two samples and cluster them together:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

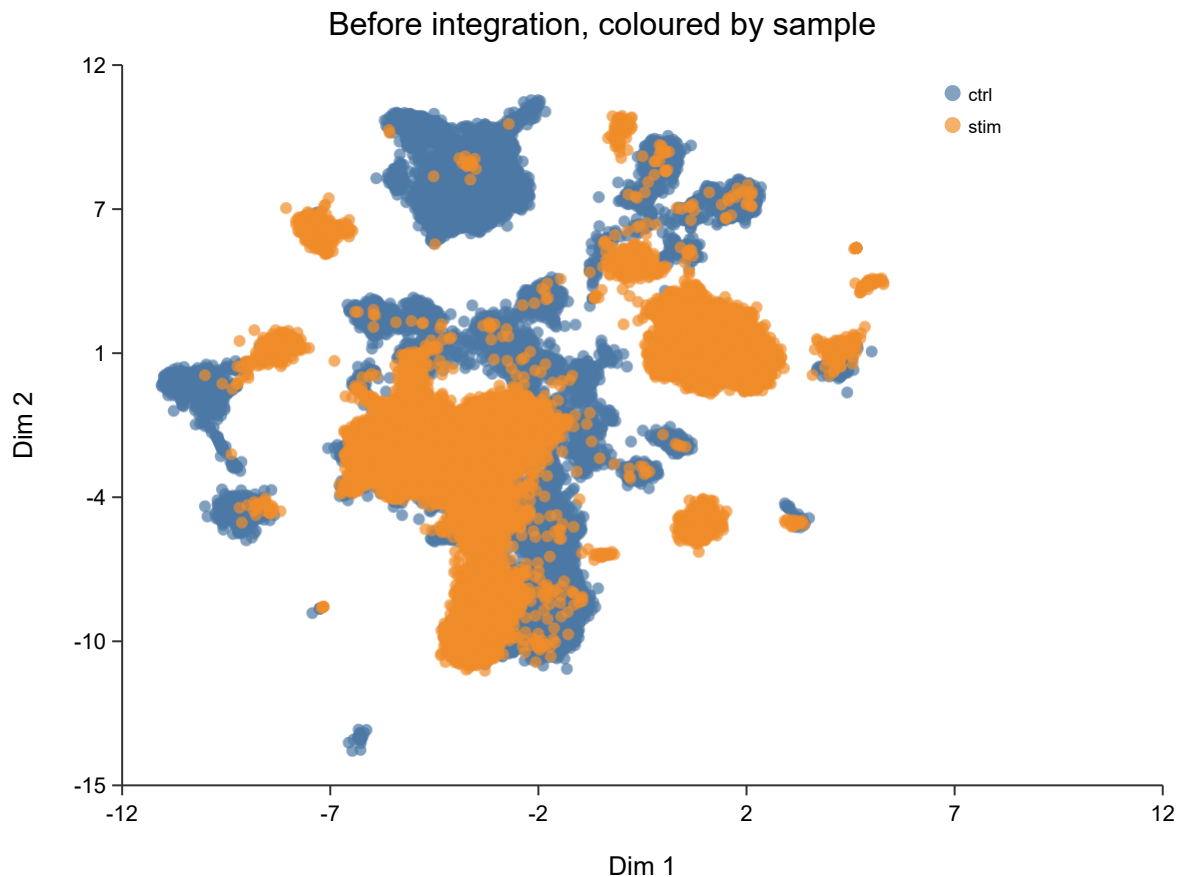
# Cells are filtered per sample; genes must match across the two for the
merge,
# so gene filtering happens after it.
let ctrl = sc.load("ctrl_raw") |> sc.filter_cells(250, 100000, 20.0)
let stim = sc.load("stim_raw") |> sc.filter_cells(250, 100000, 20.0)

let merged = sc.merge(ctrl, stim, "ctrl", "stim")
  |> sc.filter_genes(3)
  |> sc.normalize()
  |> sc.variable_genes(2000)
  |> sc.run_pca(30)
  |> sc.neighbors(15)
  |> sc.cluster_leiden(15, 0.5)

println("merged: " + str(merged.n_cells) + " cells, " +
        str(sc.get_clusters(merged) |> unique |> len) + " clusters")

let sample = merged.batch_ids |> map(|x| str(x))
write_text("integration-before.svg",
  sc.plot_embedding(merged, sample, "Before integration, coloured by
sample"))
```

```
merged: 30043 cells, 14 clusters
```



Look at the colours. Several islands are **almost entirely one sample** — a blue island beside an orange island, repeatedly. The same cell type is appearing twice, once per sample, and clustering is dutifully calling them different populations.

That is the problem integration exists to solve.

What is actually being corrected here

An honest caveat the mechanics can obscure. These two samples differ in **two** ways at once:

1. They were prepared and sequenced separately — a technical batch effect.
2. One was stimulated with interferon-beta — a real, large biological effect, and the entire point of the experiment.

Integration cannot distinguish them. Aligning the samples removes some of the interferon response along with the batch effect. That is a deliberate trade, and it is the right one *for the purpose of matching cell types across conditions*: you want the CD14 monocytes of both samples to land in one cluster so you can then ask how they differ. It would be exactly wrong to integrate and then declare the conditions similar — you removed the difference yourself.

Integrate to align cell types. Test for condition effects on the uncorrected counts, within cell type.

If your samples were biological replicates processed together and they already overlap, do not integrate at all. Cluster first; if the samples already mix, stop. Integration is a repair, not a routine step.

Harmony

Harmony (Korsunsky et al., 2019) works in PCA space rather than gene space, which is why it is fast. It alternates two steps:

1. **Soft cluster** the cells, with a penalty that rewards clusters containing a mixture of samples. A cluster that is 100% one sample is penalised, pushing toward clusters that represent cell types rather than batches.
2. **Correct within each cluster** by regressing the batch out, producing a per-cell shift.

The second step is the one that matters, and the reason is easy to miss.

Subtracting one global per-batch offset assumes the batch effect points the same way for every cell type. It usually does not — a batch can shift T cells one way and monocytes another. Correcting *within* clusters lets the correction differ per cell type, which one global offset cannot do.

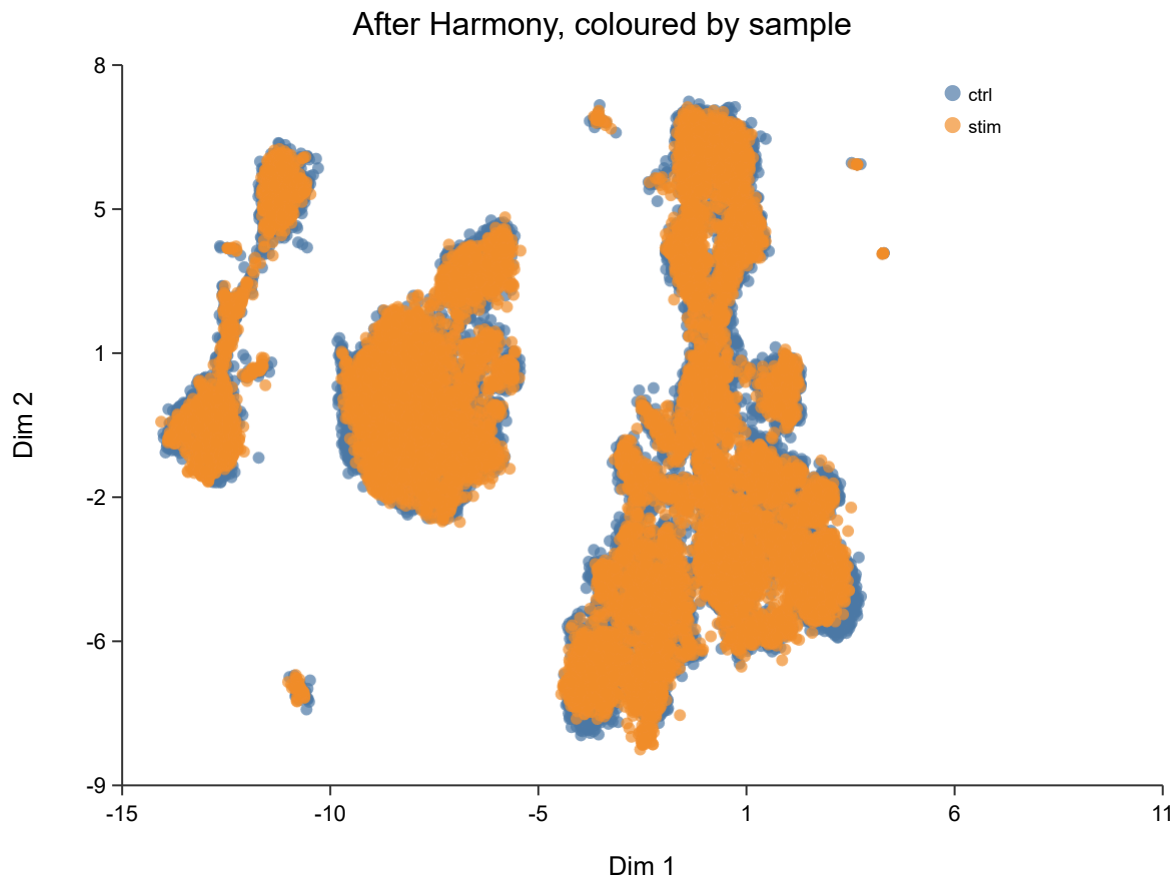
Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let ctrl = sc.load("ctrl_raw") |> sc.filter_cells(250, 100000, 20.0)
let stim = sc.load("stim_raw") |> sc.filter_cells(250, 100000, 20.0)
let merged = sc.merge(ctrl, stim, "ctrl", "stim")
  |> sc.filter_genes(3) |> sc.normalize() |> sc.variable_genes(2000)
  |> sc.run_pca(30) |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

let corrected = sc.harmony(merged, merged.batch_ids)

let sample = merged.batch_ids |> map(|x| str(x))
write_text("integration-after.svg",
  sc.plot_embedding(corrected, sample, "After Harmony, coloured by
sample"))
```

`sc.integrate` is not Harmony. It is a single global per-batch centering — a reasonable deterministic baseline, and the thing this book warns about above. Use `sc.harmony` when you mean Harmony. The two are easy to confuse because `integrate` is the more obvious name, and until recently the package exposed only the weaker one.

How to tell whether it worked

This is the part that gets skipped, and it is the part that matters.

A batch-mixing metric alone is **maximised by collapsing every cell onto a single point**. Perfect mixing, zero biology. So mixing is never sufficient evidence. Every check must come in a pair:

- **Samples mix** — each island now contains cells from both, in roughly the proportions the samples contribute.
- **Cell types stay apart** — the populations distinguishable before integration are still distinguishable after.

Compare the two figures above on both counts. The sample-specific islands are gone, and the islands themselves are still there — perhaps a dozen distinct groups rather than one cloud. Both halves hold.

If you only ever check the first, you cannot detect over-correction, which is the characteristic failure of every integration method. A UMAP that looks beautifully mixed and has quietly merged your CD4 and CD8 T cells is a worse outcome than no integration at all, because it looks like success.

The BioLang test suite for `harmony_integrate` asserts both as a pair for exactly this reason: a mutation that subtracted the regression intercept made the batches mix perfectly and collapsed cell-type separation from 15.91 to 0.00, and a mixing-only test still passed.

CCA, and why it is theory here

Canonical Correlation Analysis is the idea behind Seurat’s original integration, and it is worth understanding even though you will not run it at scale in BioLang.

PCA on either dataset alone finds the directions of greatest variance *in that dataset*, and a batch effect is often the largest single source. So PC1 becomes the batch. CCA instead looks for directions **correlated between the two datasets**. A batch effect present in only one cannot correlate with anything in the other, so it scores poorly. Shared biology — T cells behaving like T cells in both samples — scores well.

BioLang has a `cca` builtin and that property is demonstrable on small inputs. But Seurat’s CCA works on a cells $\times$ cells cross-product, and BioLang’s `Matrix::svd` is $O(n^4)$ — it stalls above roughly a hundred cells. Thirty thousand is out of the question. **The theory is runnable and the practice is not**; use Harmony, which is what lesson 09 does anyway.

Next

With cell types aligned across samples, the numbered clusters can be named: Clustering and then Markers and Annotation.

Clustering

Follows HBC lessons 10 (Clustering) and 11 (Clustering quality control).

Graph-based clustering

Three steps, and no distance threshold anywhere — which is why it suits this data.

1. **Build a k-nearest-neighbour graph.** Each cell connects to its k closest neighbours in PCA space.
2. **Weight the edges** by how much two cells' neighbourhoods overlap — the Jaccard index of their neighbour sets, $\text{shared} / (2k - \text{shared})$ — and drop any edge below $1/15$. This is a **shared** nearest-neighbour graph.
3. **Find communities** — groups more densely connected internally than externally — with Leiden or Louvain.

Step 2 is worth dwelling on. A kNN edge says “these two cells are close”; an SNN edge says “these two cells sit in the same neighbourhood”, which is a much stronger claim. It matters because in high-dimensional space the distance from a point to its nearest neighbour and to its furthest converge — so proximity alone stops discriminating exactly where single-cell data lives. Neighbourhood overlap does not degrade that way: two cells of the same type keep sharing neighbours however many dimensions you measure them in. The pruning does real work too, since chance adjacencies score low and removing them is what stops communities bleeding into each other.

`sc.neighbors` builds this by default, matching `FindNeighbors`. Pass `snn = false` for a plain kNN graph weighted by $1 / (1 + \text{distance})$, which is faster and worse; it exists so that results produced before the SNN graph existed can still be reproduced.

This page previously claimed BioLang weighted edges by neighbourhood overlap when it did not — the code used inverse distance, while the text described Seurat's behaviour and then argued, correctly, that raw distance is unreliable in high dimensions. The argument was against what the code actually did. The SNN graph was implemented afterwards, and this section now describes what runs.

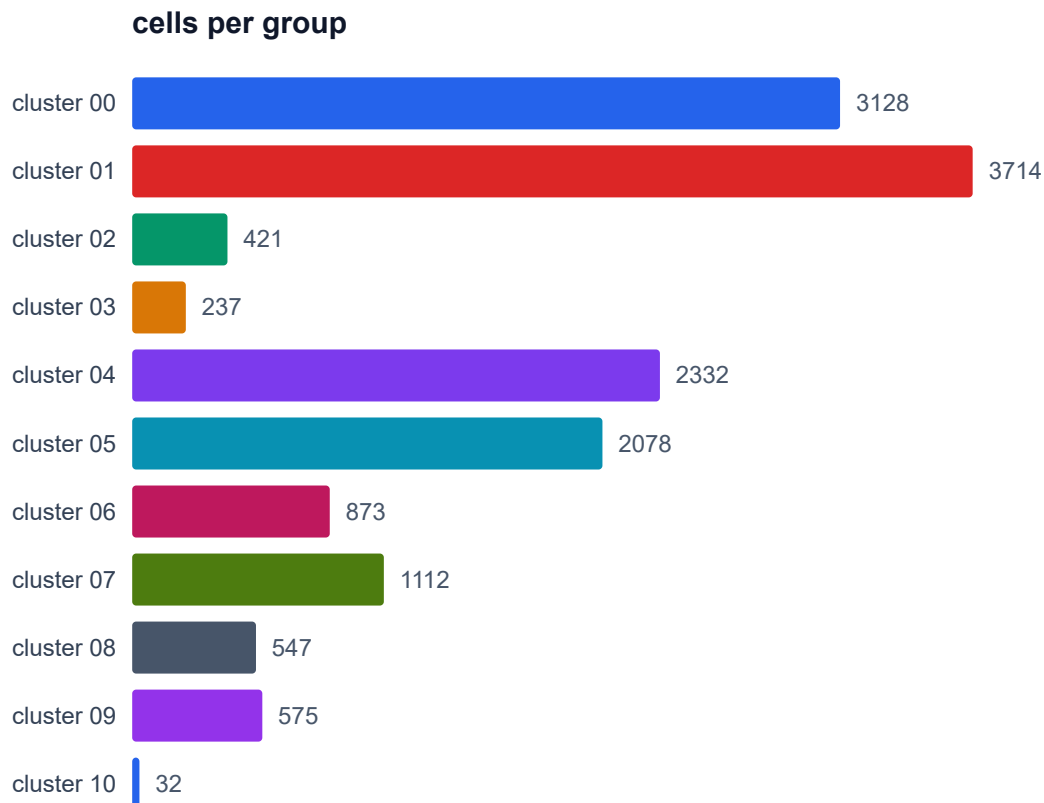
Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize()
|> sc.variable_genes(2000)
|> sc.run_pca(30)
|> sc.neighbors(15)
|> sc.cluster_leiden(15, 0.5)

let labels = sc.get_clusters(obj)
println("clusters: " + str(labels |> unique |> len))
write_text("proportions.svg", sc.plot_proportions(obj))
```

```
clusters: 11
```



Eleven clusters from 15,049 cells, ranging from 3,714 cells down to 32. That spread is normal and worth noticing: PBMCs are dominated by a few common populations, and the interesting biology is often in the small ones.

Resolution is the knob, and it is not a truth setting

resolution controls how readily the algorithm splits a community. Low values give few large clusters, high values many small ones.

There is no correct resolution. There is only one appropriate to the question. If you want broad lineages, go low. If you want to separate activation states within a lineage, go high and expect to justify each split.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let base = sc.load("ctrl_raw")
|> sc.filter_genes(3)
|> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize()
|> sc.variable_genes(2000)
|> sc.run_pca(30)
|> sc.neighbors(15)

for r in [0.2, 0.5, 1.0] {
  let c = sc.cluster_leiden(base, 15, r)
  println("resolution " + str(r) + " -> " +
    str(sc.get_clusters(c) |> unique |> len) + " clusters")
}
```

Report the resolution you used. An analysis that does not state it cannot be reproduced, because the cluster count is a function of it. The course lands on a similar count to the eleven above; do not expect an exact match, for the reasons in What Differs from the Course.

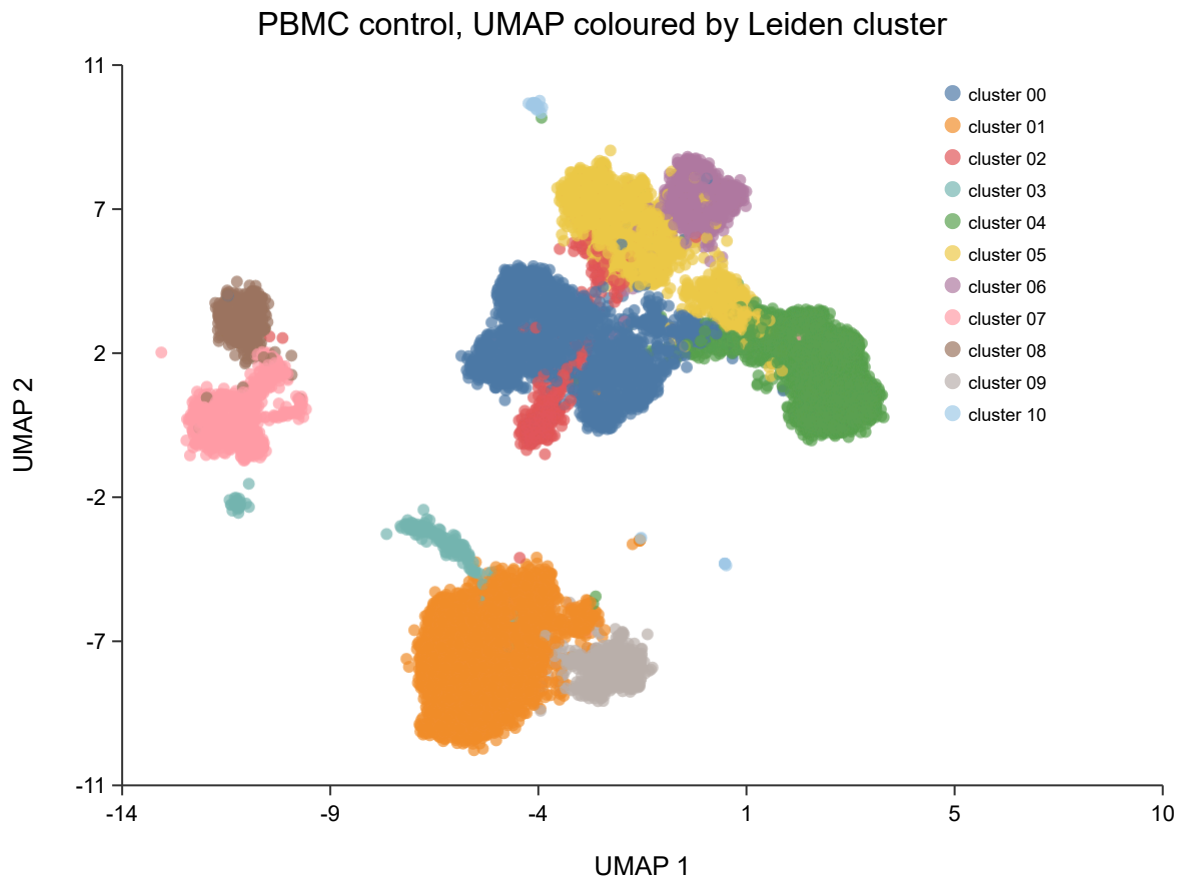
Looking at it

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
|> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

write_text("umap.svg", sc.plot_umap(obj, "PBMC control, UMAP coloured by
Leiden cluster"))
```



Read the **topology**, not the distances. Clusters 1 and 9 sit together at the bottom; clusters 7 and 8 together on the left; 0, 2, 5 and 6 form the large mass in the centre with 4 beside them. The next chapter shows those groupings are the monocytes, the B cells, and the T cells with NK adjacent — so the layout recovered relationships nobody told it about.

UMAP is for looking, not for measuring

UMAP preserves local neighbourhoods. It does **not** preserve distance between clusters. Two clusters appearing adjacent are not necessarily more similar than two at opposite ends — that spacing is partly an artifact of the layout, and it changes with the seed.

So: never argue from the picture. “These populations are related because they are next to each other” is not an argument by itself. The clustering happened in PCA space; the UMAP is a rendering of it, and every quantitative claim should come from the space, not the rendering.

This deserves more emphasis here than in most tutorials, because BioLang’s UMAP was until recently producing a single featureless blob for any input —

attraction with no reachable repulsion. The clustering was correct the whole time and the marker tables were correct the whole time; only the picture was wrong. If your embedding and your markers ever disagree, believe the markers.

The lesson most tutorials skip

The algorithm always returns clusters. Run it on pure noise and you get clusters — tidy, well-separated, meaningless. So “I have clusters” is not evidence of anything, and the course is right to give this its own lesson placed *after* clustering, because the question only becomes answerable once you have something to interrogate.

Four checks, in the order I would run them.

Is the cluster driven by QC metrics? If one cluster is just the low-UMI cells or the high-mitochondrial cells, it is a quality artifact wearing a cluster’s clothing.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
  |> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
  |> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
  |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

let d = sc.cluster_diagnostics(obj)
println("mean silhouette: " + str(round(d.mean_score, 3)) +
        " (scored " + str(d.n_scored) + " of " + str(d.n_cells) + " cells)")
```

```
mean silhouette: 0.143 (scored 2150 of 15049 cells)
```

A silhouette compares every cell against every other, which is quadratic — at 15,049 cells that is 226 million distance computations, and it does not finish in an interpreter. So it scores a deterministic subsample, the way scikit-learn’s `sample_size` does. Even subsampled this takes about three minutes; it is a diagnostic you run once, not something to put in a loop.

0.14 is low, and that is expected here rather than alarming: silhouette rewards compact well-separated blobs, and PBMC T-cell subsets genuinely grade into one another. Read it as a relative measure across resolutions, not an absolute score.

Is it driven by cell cycle? Proliferating cells of different types can cluster by phase rather than identity. `sc.cell_cycle` scores cells so you can check.

Is it one sample? With both samples merged, a cluster that is 95% one sample while the others are mixed is either an uncorrected batch effect or a genuinely condition-specific population — the second is a real finding and a much stronger claim needing much more evidence.

Is it stable? A cluster that dissolves when you nudge the resolution or the number of PCs was never robust.

`sc.cluster_stability(obj)` sweeps a set of resolutions and reports, for each, the cluster count, the mean silhouette, and the adjusted Rand index against the neighbouring resolution — a cluster set that survives a change in resolution scores high, one that dissolves scores low.

It is not run here, and the reason is worth stating rather than hiding: it computes a silhouette per resolution, and one silhouette on this sample takes about three minutes. The default five-resolution sweep is therefore a quarter-hour job. It is worth running once on a real analysis; it is not worth putting in a tutorial you want people to follow along with.

Cluster 10 above has 32 cells. Treat anything that small with suspicion until its markers say otherwise — it may be doublets, or debris, or something real and rare, and only the marker table will tell you which.

Next

Clusters are numbered, not named. Turning numbers into cell types is Markers and Annotation.

Markers and Annotation

Follows HBC lessons 12 (Seurat cheatsheet) and 13 (Marker identification).

This is the chapter the whole book exists for. Everything before it is mechanical; this is where a numbered cluster becomes a cell type.

The panel view

The fastest way to annotate PBMCs is to ask a known marker panel where it lands.

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
|> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
|> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

let panel = ["CD3D", "IL7R", "CD8A", "GNLY", "NKG7", "MS4A1", "CD79A",
             "CD14", "LYZ", "S100A8", "FCGR3A", "MS4A7", "FCER1A", "CST3", "PPBP"]

write_text("dotplot.svg",
           sc.expr_dotplot(obj, panel, "Canonical PBMC markers by cluster"))
```



Read it and the annotation falls out:

Cluster	Markers	Cell type
0, 5, 6	CD3D, IL7R	CD4 T cells
2, 10	CD3D, IL7R (weaker)	T cells
4	CD8A, GZNY , NKG7	NK / cytotoxic
7, 8	MS4A1 , CD79A	B cells
1	CD14 , LYZ , S100A8	CD14 monocytes
9	FCGR3A , MS4A7	CD16 monocytes
3	LYZ, CST3, FCER1A	dendritic cells

Now go back and look at the UMAP. Clusters 1 and 9 sit together — both monocytes. Clusters 7 and 8 sit together — both B cells. Clusters 0, 2, 5, 6 form the central mass with 4 beside them — T cells with NK adjacent. **The layout recovered the lineage relationships without being told them**, which is the strongest evidence available that the whole pipeline is working.

Why the dot plot and not a violin

The dot plot encodes two quantities at once — dot **size** is the fraction of cells detecting the gene, dot **colour** is mean expression among those cells — and that pairing is exactly what separates a real marker from an artifact.

A large pale dot means “expressed in most cells of this cluster, weakly”. A small intense dot means “a few cells, strongly”, and usually means be suspicious.

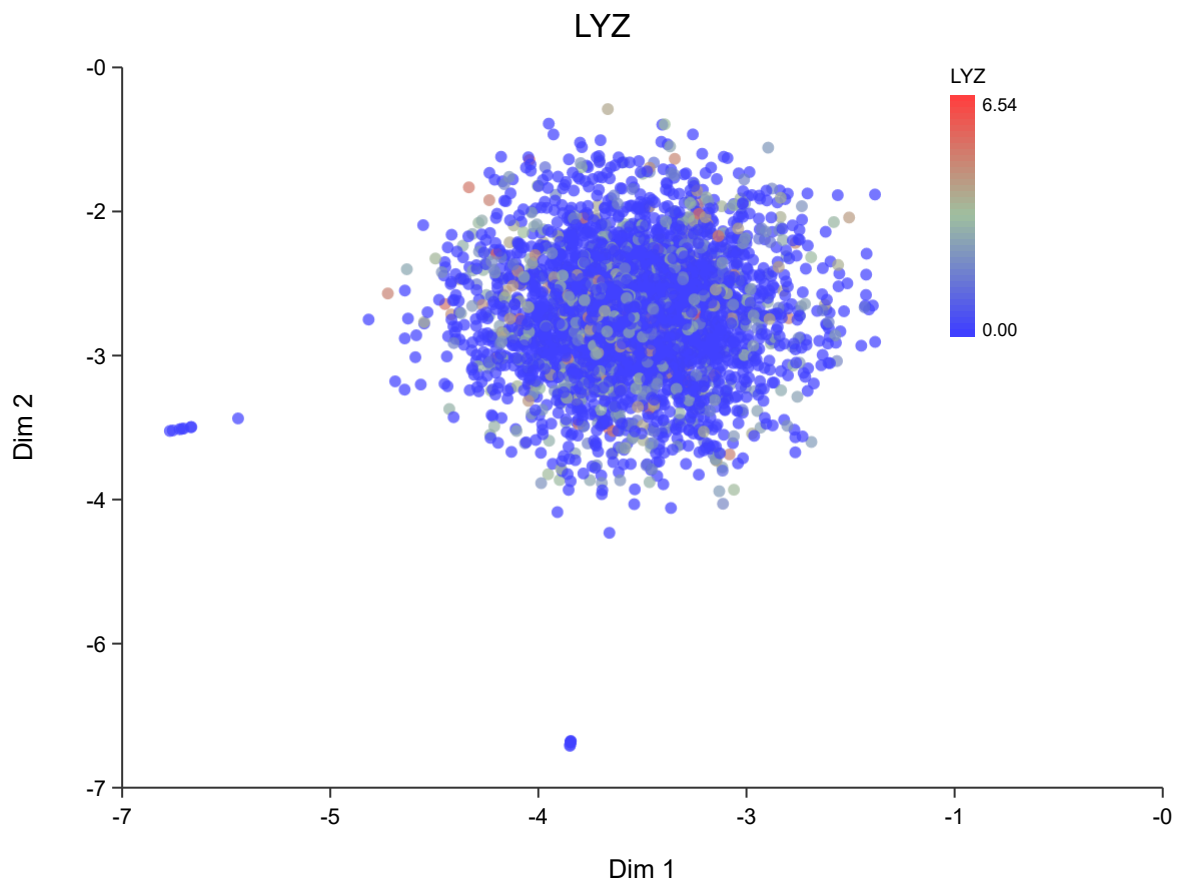
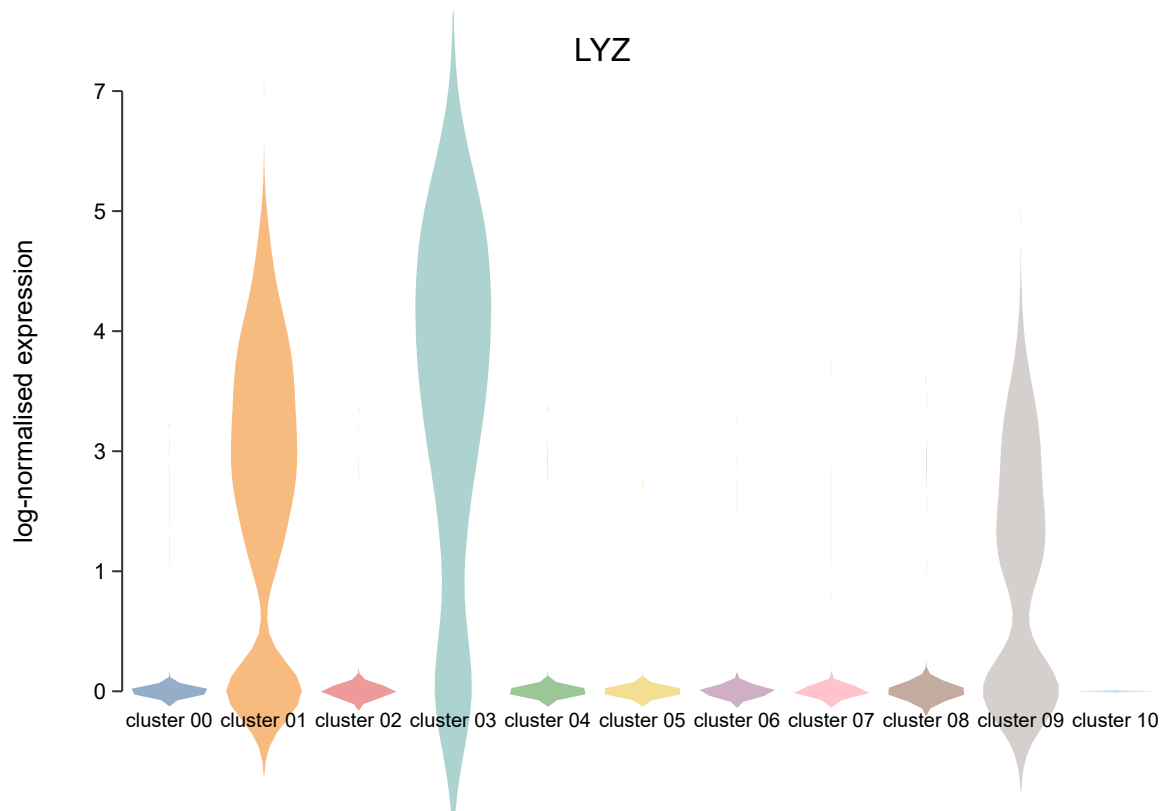
A violin hides this. Two clusters with identical violins can differ completely in what fraction of their cells express the gene at all:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
  |> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
  |> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
  |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

write_text("violin-LYZ.svg", sc.plot_violin(obj, "LYZ"))
write_text("feature-LYZ.svg", sc.plot_feature(obj, "LYZ"))
```



LYZ is high in clusters 1, 3 and 9 and near zero elsewhere — the monocytes and dendritic cells, and nothing else. The feature plot says the same thing spatially.

Finding markers you did not know to look for

The panel above assumes you already know PBMC biology. When you do not:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
  |> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
  |> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
  |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

let markers = sc.marker_table(obj, 1, 0)
println(head(markers, 8))
```

`marker_table(obj, a, b)` compares two clusters. The columns:

Column	Meaning
<code>log2fc</code>	Log2 fold change between the groups
<code>pct_a</code> , <code>pct_b</code>	Fraction of cells in each group detecting the gene
<code>pvalue</code>	Raw test p-value
<code>padj</code>	After multiple-testing correction

Read pct before you read p

The p-values here will be tiny — often reported as zero. **Do not take them seriously as evidence.**

The test asks “could these two groups have the same expression?”, and you already know the answer is no, because you *defined* the groups by clustering on this same expression data. The clustering separated the cells; the test then confirms they are separated. That is circular, and it is why marker p-values are best read as a ranking device rather than as inference. A published p-value of 10^{-300} from a marker table means “this gene ranked high”, not “this finding is certain”.

The columns carrying real information are `pct_a` and `pct_b`. A gene detected in 95% of cluster cells and 5% of the rest is a usable marker regardless of its p-value. A gene with a huge fold change detected in 10% of the cluster is not — it is a few cells with high counts, and it will not replicate.

Rank by fold change, filter by detection rate, use the p-value only to break ties.

Markers against everything else

Comparing one cluster to one other answers a narrow question. Usually you want each cluster against all remaining cells:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
  |> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
  |> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
  |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

write_text("markers.svg", sc.plot_markers(obj, 5))
```

The underlying `find_all_markers` builtin runs a Mann-Whitney U test per gene per cluster and applies **one** Benjamini-Hochberg correction across the whole table, not per cluster. Correcting within each cluster separately would understate the multiple-testing burden, since the family is every test you ran.

From markers to cell types

This step is manual, and there is no honest way around it. You match marker genes against known biology — literature, marker databases, or a reference atlas — and assign names.

Three cautions the course is right to stress:

A cluster without a clean marker set may not be a cell type. It may be doublets, stressed cells, or an over-split fragment of a real population. Cluster 10 above has 32 cells; look at its markers before giving it a name.

Markers are context-dependent. A gene marking a population in blood may mark something else in tumour. Markers from a PBMC atlas do not transfer unexamined to solid tissue — and this dataset *is* the PBMC atlas case, which is why it works so cleanly here.

Record your reasoning. “Cluster 1 = monocytes” is not reproducible. “Cluster 1: CD14+ LYZ+ S100A8+, FCGR3A–, 3714 cells — classical CD14 monocytes” is. Six

months later you will not remember, and a reviewer cannot check the first version at all.

The Seurat verbs, translated

For readers coming from the course's R:

Read this as “the step that occupies the same slot”, not “the same computation”. Several of these are approximations, and the ones that are carry a note. A table of bare equivalences would be easier to read and would mislead you about exactly the things that make numbers differ.

Seurat	BioLang	
CreateSeuratObject	sc.load / sc.from_matrix	
PercentageFeatureSet + subset	sc.qc → sc.filter_cells	filter_cells has no UMI floor or complexity term — see Quality Control
NormalizeData	sc.normalize	equivalent
SCTransform	sc.sctransform	independent implementation of the same method — see below
FindVariableFeatures	sc.variable_genes	dispersion, unbinned; Seurat's vst bins by mean first. Under SCTransform, use sc.sctransform(obj, n) instead
RunPCA	sc.run_pca	subspace iteration; Seurat uses irlba
ElbowPlot	sc.plot_elbow	
FindNeighbors	sc.neighbors	SNN with Jaccard weights, same as Seurat. Neighbour search is exact here, approximate (Annoy) there

Seurat	BioLang	
FindClusters	sc.cluster_louvain	Louvain with 10 restarts, matching <code>algorithm = 1</code> , <code>n.start = 10</code> . <code>sc.cluster_leiden</code> is the better algorithm but not the one the course ran
RunUMAP + DimPlot	sc.plot_umap	
FindMarkers	sc.marker_table	
FindAllMarkers	find_all_markers	
FeaturePlot	sc.plot_feature	
VlnPlot	sc.plot_violin	
DotPlot	sc.expr_dotplot	
RunHarmony	sc.harmony	not <code>sc.integrate</code> , which only centres each batch per gene
merge	sc.merge	

How close `sc.sctransform` is

It implements the same method: a negative binomial per gene, overdispersion estimated by maximum likelihood, then *regularized* by smoothing those estimates across genes of similar expression. The regularization is the point of the method and what the paper's title refers to. Residuals are clipped at `sqrt(n_cells/30)`, the published default.

It is an independent implementation written from Hafemeister & Satija 2019 and Choudhary & Satija 2022, not a translation — the reference is GPL-3 and BioLang is MIT.

An earlier version of this page said `sc.sctransform` used a fixed overdispersion of 100 for every gene. That was true when written. It shared the residual formula with the method and none of the regularization, which is most of it.

One known divergence remains: bandwidth for the smoothing uses Silverman's rule where the reference uses the Sheather-Jones plug-in, so the smoothed overdispersion curve differs slightly.

Next

Everything above in one script: The Whole Workflow.

The Whole Workflow

Follows HBC lesson 14 (scRNA-seq workflow).

The course closes by writing the whole analysis out in one place, and it is the most useful lesson in the set. Individual steps make sense in isolation; the shape of the thing only becomes visible end to end.

One sample, written out

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.filter_genes(3) # gene kept if seen in >= 3 cells
|> sc.filter_cells(250, 100000, 20.0) # min genes, max genes, max % mito
|> sc.normalize(10000.0) # CP10K then log1p
|> sc.variable_genes(2000) # HVGs, to focus PCA
|> sc.run_pca(30) # linear dimensionality reduction
|> sc.neighbors(15) # kNN graph, k = 15
|> sc.cluster_leiden(15, 0.5) # communities at resolution 0.5

println("cells: " + str(obj.n_cells))
println("clusters: " + str(sc.get_clusters(obj) |> unique |> len))
write_text("umap.svg", sc.plot_umap(obj, "PBMC control, UMAP by cluster"))
```

```
cells: 15049
clusters: 11
```

Nine lines, and every parameter that shapes the result is visible in them. That is the property to preserve — **the numbers are the analysis**, and hiding them inside a function makes the analysis unreviewable.

Or with the decisions printed for you

`sc.standard` runs that pipeline and reports what it did:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
|> sc.standard(nil, 2000, 15, 250, 100000, 20.0)
```

It prints the equivalent explicit pipeline, the cells in and out, and a table of every parameter marked `[set]` or `[default]` with a note on what each does — and tells you that resolution is the one to tune first.

This is **not** a black box. It prints the explicit form, so it teaches the pipeline while running it, and every decision stays inspectable in `obj.decisions`. A convenience function that concealed these numbers would produce results nobody could review; one that prints them lets you graduate to the explicit form the moment you need to deviate.

Both samples

The version this dataset is really for:

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

# Cells are filtered per sample; genes must stay identical across the two for
# the merge, so gene filtering happens after it.
let ctrl = sc.load("ctrl_raw") |> sc.filter_cells(250, 100000, 20.0)
let stim = sc.load("stim_raw") |> sc.filter_cells(250, 100000, 20.0)

let merged = sc.merge(ctrl, stim, "ctrl", "stim")
|> sc.filter_genes(3)
|> sc.normalize()
|> sc.variable_genes(2000)
|> sc.run_pca(30)
|> sc.neighbors(15)
|> sc.cluster_leiden(15, 0.5)

println("merged: " + str(merged.n_cells) + " cells, " +
        str(sc.get_clusters(merged) |> unique |> len) + " clusters")

let corrected = sc.integrate(merged, merged.batch_ids)
```

```
merged: 30043 cells, 14 clusters
```

The ordering is not cosmetic. `filter_genes` before the merge leaves the two samples with different gene sets, and `sc.merge` refuses:

`sc_merge_objects()` requires identical genes in identical order.

Refusing is the right behaviour — quietly aligning by position would produce a matrix where column 400 means one gene in one half and another gene in the other.

Checking it

Requires the CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("ctrl_raw")
  |> sc.filter_genes(3) |> sc.filter_cells(250, 100000, 20.0)
  |> sc.normalize() |> sc.variable_genes(2000) |> sc.run_pca(30)
  |> sc.neighbors(15) |> sc.cluster_leiden(15, 0.5)

println(sc.cluster_diagnostics(obj))
println(sc.cluster_stability(obj))
```

On real data you have no ground truth, so the diagnostics matter more, not less. The checks that carried weight in this book were the ones where two independent views had to agree: the marker dot plot and the UMAP topology both say clusters 1 and 9 are monocytes and 7 and 8 are B cells. Neither was told about the other.

What the workflow does not include

Being explicit about the boundary, because a pipeline that runs to completion invites the belief that it is finished.

- **Annotation is not in it.** The clusters are numbered. Naming them is manual and needs biology the data cannot supply — see Markers and Annotation.
- **Differential expression between conditions is not in it.** With ctrl and stim in hand it is the obvious next question, and it needs a pseudobulk design with replicates. Testing across cells treats cells from one sample as independent replicates, which they are not, and produces confidently wrong p-values.
- **Doublet detection is not in it.** `sc.doublers` and `sc.flag_doublers` exist; add them if your loading concentration was high.
- **Cell cycle regression is not in it.** `sc.cell_cycle` scores cells so you can check whether it matters before deciding to remove it.

Timings

Measured on a laptop, single-threaded, on this dataset:

Step	
Load one raw matrix (737,280 × 33,538)	2 s
Filter to 15,049 cells	5 s
Normalize → HVG → PCA → neighbours → cluster	15 s
UMAP and plot	25 s
Both samples merged (30,043 cells)	40 s
Harmony on 30,043 cells	minutes

Nothing here needs a cluster, but the two-sample chapters are not instant.

Where to go next

Read the course itself, at <https://hbctraining.github.io/Intro-to-scRNAseq/>. It teaches the same material in R with Seurat, which remains the implementation everything else is measured against. Credit and licence are on Attribution and Licence, and the places BioLang and Seurat part company are in What Differs from the Course.

What differs from the course

A companion that claims to match and then quietly substitutes something weaker is worse than one that lists its gaps. Here they are.

Current state, stated plainly. BioLang's single-cell pipeline was substantially rewritten to match the published Seurat methods: the PCA (which was returning components out of order — a real defect), the neighbour graph (now SNN with Jaccard weights), Louvain (now with the reference's ten restarts), and normalization (now genuinely regularized negative binomial rather than a fixed overdispersion).

On the course data, the historical Harmony example reports **18 clusters with GPU auto-detection and 16 with `--no-gpu`**. The final corrected CCA validation notebook now reports **21 on the measured CPU backend**. A current GPU CCA run has not been measured after the latest SCTransform conformance changes. The historical rendered HBC object has 17 clusters; an independent current Seurat CCA run described below has 19.

The numbers quoted further down this page — 17, then 16 — come from earlier stages of that work and are kept because the *reasoning* about them is still the point. But do not read them as current output.

The honest implementation trajectory was 17, 16, 25, 20, 21, then **18 on GPU / 16 on CPU**, followed by 18 and now **21 for the final CPU CCA workflow**. That is evidence a single cluster count was never a sound target. The independent Seurat run now gives a real comparison: all 29,629 cells join exactly, but the corrected BioLang CPU CCA run versus Seurat has ARI 0.5750 and 69.92% optimally mapped-cell accuracy. Agreement claims below are based on those cell-level measurements, not on matching an integer.

The lesson map

Fourteen lessons, seven chapters. Course lessons that exist to set up an R environment have no BioLang counterpart and are folded into their neighbours.

HBC lesson	Covered in	Status
01 Intro to scRNA-seq	The Biology and the Matrix	Full
02 Generation of the count matrix	The Biology and the Matrix	Read-only
03 Quality control setup	Quality Control	Folded in
04 Cell Ranger QC	Quality Control	Partial
05 Quality control	Quality Control	Full
06 Theory of PCA	Normalization and PCA	Full
07 SCTransform	Normalization and PCA	Full
08 Integration: CCA theory	Integration	Full workflow, approximate numerics
09 Integration: Harmony	Integration	Full
10 Clustering	Clustering	Full
11 Clustering quality control	Clustering	Full
12 Seurat cheatsheet	Markers and Annotation	Translated
13 Marker identification	Markers and Annotation	Full
14 The whole workflow	The Whole Workflow	Full

The numbers will not match exactly

Same data, same thresholds, different implementations. Expect agreement on **shape** and disagreement on **digits**.

The pipelines are not identical, and the differences are specific:

	This book	The course
QC filter	gene floor + mito cap	+ UMI floor + complexity

	This book	The course
Cells kept (ctrl)	15,049	14,847 with their full filter
Normalization	<code>normalize</code> (CP10K + log1p)	SCTransform
PCs	30	40
Clustered on	one sample at a time	the integrated object

The course also clusters at **resolution 0.8**, where this book uses 0.5.

Can you align them? Close — and this is where an earlier claim broke

None of those differences are limitations. BioLang has `sc.sctransform`, takes any PC count, and the filter can be written by hand. Matching the course's dimensions and filters while using its documented Harmony alternative — both samples, their four-criterion filter, SCTransform, 40 PCs, resolution 0.8 — gives:

```

mode      cells  clusters  wall time
GPU auto  29629    18        71.9 s
--no-gpu  29629    16        72.5 s
HBC HTML           17

```

The course's marker lesson assigns identities to clusters 0 through 16: 17 clusters. That is a historical output, not a timeless Seurat truth. The independent HBC CCA workflow run with Seurat 5.5.1 produced 19 clusters at the same selected resolution. Large-data neighbour ranking deliberately uses different BioLang GPU and CPU backends, so cells near a graph boundary can move; the run header now prints the selected backend before analysis.

The executable CCA validation notebook is the direct ground-truth comparison. The final CPU run produced 21 clusters versus Seurat's 19. Against Seurat it has ARI 0.5750, adjusted mutual information 0.7269, and 69.92% one-to-one mapped accuracy. The shared variable-feature set is 2,716 of 3,000 genes (90.53%). Its approximate integrated-PC 15-neighbor Jaccard is 0.2278. Those measurements establish partial agreement, not numerical parity. The GPU CCA backend was not rerun after the latest SCTransform conformance changes, so its earlier figures remain historical rather than current comparison results.

The standalone transform has the same important distinction. On the HBC control, `log10(theta)` Pearson correlation is 0.999216, but raw theta has slope

0.9389 and median relative error 7.26%. A top-feature residual probe spanning the 3,000 highest-variance genes is much stronger: residual correlation is 0.999824, slope is 0.9953, and RMSE is 1.92% of the oracle residual SD. The scale-sensitive validation therefore passes for these residuals but not for theta; this book does not call that 99% SCTransform parity.

Marker agreement is lower: 2,993 mapped cluster/gene pairs overlap, 191 of the mapped top-50 pairs overlap, and 7 of 15 canonical marker genes peak in the same mapped cluster. These are useful diagnostics, but none is evidence for a 95% equivalence claim.

Current Seurat/BioLang resource benchmark

Both programs were run independently on the same Windows host and the same HBC inputs. Peak memory is the process `PeakWorkingSet64` counter; it is not R's internal garbage-collector estimate and it does not add dedicated GPU memory.

Program	Wall time	Peak working set	Result
BioLang release, CPU	481.2 s	16.06 GiB	21 clusters
Seurat 5.5.1 / R 4.5.2	1548.3 s	12.50 GiB	19 clusters

On this measured pair BioLang is 3.22 times faster, but uses 1.28 times as much peak host memory. Before stage-local notebook lifetimes and native compact-matrix integration, the BioLang workflow peaked at 20.25 GiB. Earlier runs took 394.0 seconds for BioLang and 909.3 seconds for Seurat, so runtime varies substantially; the direction did not change. The requested state is faster **and** lower-memory than Seurat. Only the first half is currently demonstrated.

Why older versions of this page said 17, then 16

Not wrong in the sense of a mistyped number — the run really did print 17. It was wrong in the sense that mattered: **the 17 depended on selecting variable genes with a method that does not belong on these values**, and it stopped being 17 the moment that was corrected.

The pipeline was `sctransform()` followed by `variable_genes(3000)`. `variable_genes` ranks by dispersion — variance divided by squared mean — which is the standard heuristic for log-normalized counts. Pearson residuals are not log-normalized counts. They are *centred*: `mu` is fitted to the row and column margins, so each gene's mean residual sits near zero by construction. Dividing by

the square of a near-zero number does not rank genes by variability; it ranks them by how close their mean landed to zero, which is arithmetic noise.

`sc.sctransform(3000)` now selects on **residual variance**, which is what the sctransform paper proposes and what Seurat's `SCTransform` returns.

The following is a historical controlled comparison. Both rows were run against the same earlier binary, on the same cells, with everything downstream held fixed — same PCA, same Harmony, same Leiden, same resolution. Only the ranking differed:

Gene selection on the residuals	Clusters	Time	Peak memory
Dispersion — variance ÷ mean ²	17	247 s	16.8 GB
Residual variance — the paper's	16	206 s	6.3 GB

That experiment established why the 17 was not valid evidence of agreement: the defensible method gave 16 on that binary, while the wrong method happened to give 17. **Getting the reference's number out of a method the reference does not use is not agreement.** Later PCA, SNN, and SCTransform changes moved the current measured result to 18; the historical table is retained as evidence for the one-variable experiment, not presented as today's GPU output. Its 16 is also not the current CPU result by provenance, even though the number happens to be the same.

Note that the wrong method is the slower and hungrier one too. Nothing about the 17 was cheap.

The current mismatch is real. Candidates are the HVG overlap, the integration approximation, PCA, neighbour-search backend, SNN details, and optimizer tie-breaking, all of which differ; which accounts for it is not something the cluster count alone can say.

For contrast, on the control sample alone:

```
cells: 14847
SCTransform / 40 PCs / res 0.8    GPU auto    --no-gpu
log1p / 30 PCs / res 0.5          15          16
                                   8           8
```

Same cells, same code, two parameter sets. The difference was the pipeline, not the implementation. These values were rerun after SNN became the default; the older 12/10 values were pre-SNN and are no longer presented as current output.

`aligned.bl` in Downloads runs the single-sample comparison; `exact.bl` runs the full integrated configuration. On the measured release build, GPU-auto runs took 34.9 and 71.9 seconds respectively; CPU-only runs took 33.0 and 72.5 seconds.

Runtime regression recheck

An earlier integrated run regressed from 23 to 69 minutes, with the PCA's 300-sweep ceiling and a near-degenerate spectrum suspected but not measured. That concern was rerun rather than silently retired. On 9 August 2026, `exact.bl` completed in 71.9 seconds wall time and reproduced 18 clusters. An instrumented second run completed in 70.4 seconds with this breakdown:

Stage	Time
Load and QC	12.1 s
Merge and gene filter	0.9 s
SCTransform	12.3 s
PCA	13.3 s
Harmony	27.4 s
Neighbours	2.4 s
Leiden	2.0 s

This was the release build with GPU auto-detection selecting an NVIDIA RTX 3080. PCA is not the dominant stage on this dataset. For inputs above 5,000 cells, its current implementation fits on 5,000 deterministic rows and uses at most six subspace sweeps; the `MAX_SWEEPS = 300` path applies only to smaller inputs. Its convergence check follows Rayleigh quotients, so free rotation of basis vectors inside a near-degenerate subspace does not itself prevent convergence. The old 69-minute regression therefore does not reproduce in the current pipeline, while the explicit record here keeps it from disappearing as an unexamined assumption. A separate `--no-gpu` run completed in 72.5 seconds and produced 16 clusters; this backend-sensitive result is documented rather than folded into a single inferred number.

What this cost to make possible

Three memory problems, in the order they surfaced.

The integrated run died outright. `sc_sctransform` was paying for its output three times: a dense copy of the sparse input, a second array for the residuals, and then every element boxed into a `Value` inside nested lists — about $4 + 4 + 12$ GB. Pearson residuals are dense by construction, so 4 GB is real; the other 16 were not. It now streams the sparse input into one flat array and returns a matrix.

It then died more politely, at 3.95 GB. Residuals were still being computed for all 16,681 genes when the next step keeps 3,000 and discards the rest.

`sc.sctransform(3000)` ranks genes by residual variance and materialises only those, which is what `SCTransform(variable.features.n = ...)` does.

Measured on this pipeline: **16.8 GB peak uncapped, 6.3 GB capped**, and 247 s down to 206 s.

And the 6.3 GB was still too much. `Value::Matrix` held its matrix inline while `Value::SparseMatrix` had already been moved behind an `Arc`, so every record spread and every pipeline stage deep-copied the whole thing. Peak was 9.0 GB before that fix and 6.3 GB after, with the result unchanged. It was the same bug, in the same file, as the one that had made reading a single gene take seven minutes — just on the other matrix type.

So the honest historical sequence is: the numbers did not match; three separate memory bugs were in the way; fixing them let the controlled comparison run; and that comparison said 16, not the 17 the page had been claiming. The separately rerun current pipeline now says 18, as recorded above.

What still will not match

Cluster **numbering** is arbitrary in both — cluster 7 here is not cluster 7 there. Beyond that, HVG selection differs, the PCA is a different implementation, and Leiden breaks ties differently, so the boundaries between adjacent clusters will not be identical cell for cell.

And note what the 17 episode above demonstrates: **a matching cluster count was never strong evidence of a matching partition.** One number agreeing between two pipelines with different gene selection, different PCA and different community detection is a weak signal, and it turned out here to be a coincidence produced by a method error. Treat count agreement as a sanity check that nothing is catastrophically wrong, not as validation.

This book's chapters keep the simpler settings, because log-normalization is easier to explain and fifteen seconds beats three and a half minutes while you are learning the shape of the workflow. That is a teaching choice, and now an explicit one with the cost measured.

The three real gaps

Lesson 02 — generating the count matrix. This covers Cell Ranger: demultiplexing, barcode correction, alignment, UMI collapsing. BioLang does none of it, and neither does Seurat — it is upstream of both. BioLang starts where Cell Ranger stops. The chapter explains what happened upstream, because you cannot interpret a UMI count without knowing what a UMI is, but it does not pretend to run it.

Lesson 04 — Cell Ranger’s own QC report. The course reads the `web_summary.html` Cell Ranger emits. BioLang has no parser for it. Nearly all of the same quantities can be recomputed from the matrix, and the QC chapter does — but **sequencing saturation** and **fraction of reads mapped to the transcriptome** are properties of the reads, and the reads are gone by the time you have a matrix. If you have the file, read it in Cell Ranger’s viewer.

Lesson 08 — exact CCA parity at realistic scale. BioLang now runs the full 29,629-cell CCA workflow with a bounded CountSketch subspace method and does not materialise the cells-by-cells cross-product. That makes the practice runnable, but it is a numerical approximation rather than Seurat’s IRLBA/Annoy dependency path. The measured ARI and neighbourhood overlap above show that scalability is implemented while 95% result parity is not.

Things this book found that the course cannot tell you

Running the course’s data through a different implementation is a good way to find bugs in the implementation. Several turned up while this book was written, and they are worth knowing about because they shape what you should trust:

- **UMAP produced a featureless blob for any input.** The layout had attraction but no reachable repulsion. Fixed, with tests. If you are running an older build and your embedding looks like a single cloud, this is why — and note that the clustering and marker tables were correct throughout. **If your embedding and your markers disagree, believe the markers.**
- **Reading one gene across 15,000 cells took seven minutes**, because the sparse count matrix was deep-copied on every value clone. Fixed; it is now free.
- **Plotting the raw droplets never finished**, because the scatter path emitted one vector circle per point and accumulated them quadratically. Large scatters are now rasterised.

- **Violin plots crashed on unevenly sized clusters.** The padding value for ragged columns was `-inf`, which is not valid JSON. An evenly sized synthetic fixture never triggered it; real clusters ranging from 3,714 cells to 32 do immediately.
- **Dense matrices were deep-copied on every clone**, exactly as the sparse ones had been before that was fixed. Worth 2.7 GB on this pipeline. If you fix a bug of this kind, check whether its sibling has it too — here, nobody did for months.
- **`sc.sctransform` ignored its own second argument**, because the package facade forwarded one argument to a function that now took two. It was silent because **BioLang discards extra arguments to a user-defined function without complaining** — `f(1, 2, 3)` on a one-parameter `f` returns quietly. That is a language-level defect and it is still open; until it is fixed, a typo in an argument list is invisible.

The last two are the general lesson, and it is not the one about synthetic data. Synthetic data tests the code you wrote and real data tests the assumptions you did not know you had made — but both of those bugs were found by *running the thing and watching a number that should not have moved*. Neither a test suite nor a bigger dataset would have surfaced them.

Downloads

None of the code in this book runs in the browser: every example imports the `singlecell` package and most read or write files, and package imports and file I/O are CLI-only. So the pages show no Run button — copy the code, or take one of these.

The starter kit

singlecell-starter.zip — the package, the data-download script, every chapter script and the notebook.

```
unzip singlecell-starter.zip && cd singlecell-starter
bl install ./singlecell
python get-data.py
bl run ch01.bl
```

Full setup, including installing BioLang itself, is in Getting the Data.

The data script

get-data.py — fetches the two 10x matrices from the HBC course's own host. 3.2 GB down, ~90 MB kept; the archive is deleted unless you pass `--keep`. Also included in the kit.

Chapter scripts

One per chapter, each self-contained.

Chapter	Script	Blocks
The Biology and the Matrix	ch01.bl	1
Quality Control	ch02.bl	3
Normalization and PCA	ch03.bl	4

Chapter	Script	Blocks
Integration	ch04.bl	2
Clustering	ch05.bl	4
Markers and Annotation	ch06.bl	4
The Whole Workflow	ch07.bl	4

These are not quick, and here is what they actually cost. Measured on a laptop, single-threaded:

Script	Time
ch01	2 s
ch02	~1 min
ch03	~2 min
ch04	3 min 35 s
ch05	3 min 50 s
ch06	over 9 min
ch07	over 9 min

Two things drive this. Each block on a page is written to stand alone, so a reader can copy any one without having run the others — which means a concatenated chapter rebuilds the pipeline from the raw matrix once per block. And two operations are inherently expensive on 15,000 cells:

`plot_markers` runs a test per gene per cluster and takes **5 minutes** by itself, and `cluster_diagnostics` computes a silhouette and takes **3**.

`ch06` and `ch07` were confirmed to run only in pieces, not as single scripts inside a ten-minute budget. Nothing in them is broken — every block produced the figure on its page — but if you want to follow along interactively, lift a single block rather than running a whole chapter.

The course-aligned pipeline

aligned.bl — the course's settings rather than this book's: their four-criterion QC filter, SCTransform, 40 PCs, resolution 0.8. It prints both cluster counts side by

side so you can see what the parameters cost:

```
cells: 14847
SCTransform / 40 PCs / res 0.8      GPU auto      --no-gpu
log1p / 30 PCs / res 0.5             15             16
                                      8              8
```

Measured after the SNN-default change on 9 August 2026: 34.9 seconds with GPU auto-detection enabled (NVIDIA RTX 3080), and 33.0 seconds with `--no-gpu`.

exact.bl — the full integrated configuration: both samples, their filter, SCTransform, Harmony, 40 PCs, resolution 0.8:

```
mode      cells  clusters  wall time
GPU auto  29629    18       71.9 s
--no-gpu  29629    16       72.5 s
Historical HBC HTML      17
```

Both modes were measured on the same release build. Large-data neighbour ranking uses different numerical/search backends on GPU and CPU, so cells near a graph boundary can move and the Leiden count is not guaranteed to match. An independent current HBC CCA run with Seurat 5.5.1 gives 19 clusters; the cell-level comparison is recorded in What Differs from the Course. Harmony was the largest timed stage, not PCA or neighbour search; the measured breakdown is recorded in What Differs from the Course.

What Differs from the Course has the full account: what was rewritten to match Seurat's published methods, why the count moved *away* from 17 as those fixes landed, and why one cluster count was a poor target to aim at in the first place.

The notebook

hbc-scrnaseq.bl — prose and code together, runnable end to end:

```
bl notebook hbc-scrnaseq.bl
```

Several chapters write `.svg` figures into the current directory. The notebook carries the same content as the book, minus the figures, whose paths would not resolve beside a downloaded file.

What you cannot download here

The dataset itself. It belongs to the study authors and is hosted by the HBC training team; `get-data.py` fetches it from them rather than redistributing a copy. See Attribution and Licence.